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RESEARCH PROPOSAL
CLIMB-AI: Explainable Prediction of Compound Climate-Related Pedestrian Mobility Disruption on Dhaka Streets

Research Topic

Explainable machine learning for street-level heat, monsoon-waterlogging, and pedestrian-mobility risk in Dhaka, Bangladesh

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CLIMB-AI: Explainable Prediction of Compound Climate-Related Pedestrian Mobility Disruption on Dhaka Streets

1. RESEARCH PROBLEM AND BACKGROUND

Pedestrians in Dhaka can face extreme heat, monsoon waterlogging, and difficult street conditions at the same time, even though these problems are often studied separately. Dhaka heat studies have linked land-surface temperature to urban growth, vegetation, and built form [1]-[4], while household-level research documents substantial disruption from monsoon waterlogging [5]. International pedestrian studies also show that heat can influence route choice and reduce effective accessibility [6], [7].

Street conditions may change how strongly these hazards affect walking. Two streets with similar temperatures can have very different pedestrian conditions if one has shade, drainage, and a usable footway while the other is exposed, obstructed, or waterlogged. This study therefore uses the street segment as the unit of analysis and tests whether heat exposure, waterlogging susceptibility, and street conditions together predict an independently observed change in pedestrian activity better than simpler single-hazard models.

The primary outcome will be Pedestrian Flow Suppression (PFS), calculated from matched field counts, with a separate physical-passability audit as supporting evidence. Keeping the outcome independent from the predictor variables avoids constructing a circular risk index and makes the central comparison directly testable.

2. EXISTING LITERATURE AND RESEARCH GAP

Existing research provides the main components of the proposed framework. Dhaka and Bangladesh studies have examined urban heat using Landsat, environmental indices, and machine learning [1]-[4]. Waterlogging research has documented recurring losses and transport disruption in Dhaka [5]. Pedestrian studies outside Dhaka have shown that heat affects route choice, walking reach, and access to destinations [6]-[8]. Explainable machine-learning studies also demonstrate how tree-based models and feature-attribution methods can be used to interpret urban climate risk [9].

The remaining question is whether these components add useful predictive information when combined at street level and tested against an independently observed pedestrian outcome in Dhaka. The literature reviewed for this proposal did not identify a study using this exact combination of selected Dhaka street segments, compound heat and waterlogging information, repeated field-observed pedestrian disruption, spatially blocked validation, local explainability, and intervention-sensitivity analysis. CLIMB-AI therefore focuses on that specific test rather than claiming that its individual methods are new.

3. RESEARCH QUESTION AND HYPOTHESES

Primary research question: Can a model combining thermal exposure, waterlogging susceptibility, and street-level conditions predict weather-related pedestrian mobility disruption in selected Dhaka neighbourhoods more accurately than heat-only, waterlogging-only, and fixed additive baselines?

Secondary objective: Among street segments predicted to be at high risk, identify which modifiable street features are associated with the largest model-predicted changes in disruption.

H1: Under spatially blocked cross-validation, the compound model will outperform the strongest heat-only, waterlogging-only, and fixed additive baseline in predicting PFS.

H2: Compound high-risk streets will show greater PFS and worse passability during adverse conditions than lower-risk streets observed under comparable schedules.

H3: Sensitivity analysis will identify different potentially useful street improvements across different high-risk street types rather than one universally dominant intervention.

4. PROPOSED METHODOLOGY

4.1 Study Area, Sampling, and Unit of Analysis

Approximately 90 street segments will be sampled across three contrasting Dhaka neighbourhoods selected for variation in vegetation, built density, street form, and known waterlogging conditions, while also considering researcher access and safety. A segment will normally extend about 50-150 m between meaningful OpenStreetMap network nodes [16]. The study is a selected-neighbourhood pilot rather than a citywide survey, allowing repeated field observations while keeping the scope realistic for a student project.

4.2 Predictor Variables and Data Sources

Predictors will follow a hazard-exposure-vulnerability structure consistent with IPCC climate-risk framing [10]. Thermal exposure will use USGS Landsat 8-9 Collection 2 Level-2 surface-temperature products [11], supported by field temperature and relative-humidity observations where equipment is available. Vegetation and built form will be represented by Sentinel-2 Level-2A and Landsat surface-reflectance indices such as NDVI and NDBI [12]. SRTM elevation and slope will provide broad terrain context [13]. CHIRPS v3 or an available local rain gauge will describe city-scale rainfall forcing [14]; CHIRPS will not be treated as street-scale rainfall. GHSL will provide population context [15], and OpenStreetMap will provide the street network and destination context [16].

Street-level variables will be observed directly and will include shade, drainage condition, usable footway width, continuity, barriers, and standing water. The predictor set will be specified before modelling and kept deliberately small; strongly redundant variables will be consolidated to reduce overfitting in the pilot sample. PM2.5 will only be added as a street-level predictor if a defensible fine-resolution exposure source becomes available; otherwise it will remain contextual [17].

4.3 Independent Outcome and Field Observation

Each segment will be observed under three broadly matched conditions: a comparatively benign dry baseline, a hot/dry condition, and a post-rain condition. Visits will be matched as closely as possible by weekday type and time of day to reduce confounding from school, commuting, and market schedules. Each visit will include a 10-minute pedestrian count and a standardized street audit. The audit will record shade, usable footway width, drainage, barriers, standing water, and passability on a four-level scale: 0 = normal passage; 1 = minor slowing; 2 = substantial obstruction or detour; 3 = ordinary passage effectively blocked or unsafe. At least 20% of audit observations will be independently double-rated to check scoring agreement.

The primary continuous outcome will be PFS = ln(1 + B) - ln(1 + A), where B is the matched baseline pedestrian count and A is the adverse-condition count. Positive PFS indicates lower observed pedestrian activity during the adverse condition. PFS will be treated as a mobility-disruption proxy, while the separate passability score will help distinguish reduced activity from physical obstruction.

4.4 Modelling and Spatial Validation

Simple models will be tested first: a linear model, a heat-only model, a waterlogging-only model, and a fixed additive multi-hazard index. These will be compared with Random Forest and XGBoost. XGBoost is designed for nonlinear interactions in tabular data [18], but it will only be retained if it improves performance on spatially held-out data. More complex graph models are outside the scope of this pilot because the planned sample size does not justify the additional model complexity.

Nearby streets share environmental characteristics, so randomly splitting individual observations could produce an overly optimistic estimate of performance. Street segments

will therefore be assigned to geographic blocks for cross-validation, following guidance for spatially structured data [19]. Repeated observations from the same street will remain within the same fold, and hyperparameter tuning will occur only inside the training data. PFS will be evaluated primarily with MAE and RMSE, with R^2 reported secondarily. Passability classification will use macro-F1 and balanced accuracy. H1 will only be supported if the compound model improves held-out performance relative to the strongest simpler baseline.

4.5 Explainability and Intervention Sensitivity

18 SHAP will be applied to the best validated tree-based model to examine global feature importance and the features contributing to individual street predictions [20]. SHAP will explain the fitted model rather than establish causation. For selected high-risk streets, a secondary sensitivity analysis will vary only potentially modifiable features such as shade, vegetation, drainage condition, usable footway width, or barrier score while holding other predictors constant. The resulting change in prediction will be reported as model sensitivity to the scenario, not as the expected causal effect of physically implementing the intervention.

5. EXPECTED RESULTS, SIGNIFICANCE, AND LIMITATIONS

The study will test whether combining climate exposure and street conditions provides useful additional information about pedestrian disruption in the sampled neighbourhoods. Expected outputs include spatial predictions for the sampled street network, direct comparison with simpler baselines, field passability observations, and SHAP-based explanations. A negative result would also be informative: if the compound model does not outperform simpler baselines, the added complexity would not be justified for this pilot dataset.

If the model performs well, the results could help identify streets that deserve closer investigation for shade, drainage, footway, or accessibility improvements. The study will not claim to produce a citywide Dhaka risk map from approximately 90 streets. Instead, it tests a method that could later be expanded or recalibrated with a larger dataset.

Important limitations include cloud-related gaps in satellite observations, the difference between land-surface temperature and human thermal stress, the coarse scale of SRTM and CHIRPS relative to individual streets, incomplete OpenStreetMap information, and non-weather influences on pedestrian counts. Matching observation schedules, field measurements, passability audits, a constrained predictor set, and spatial validation will reduce but cannot eliminate these limitations. No names or sensitive pedestrian information will be collected; identifiable pedestrian imagery will not be retained; and field observations will not require entering floodwater or unsafe traffic conditions.

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